

PEYTON RAMSEY

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EDUCATION

University of New Hampshire, Durham, NH | MS in Business Analytics (MSBA), GPA: 3.63 | August 2025 – May 2026

Graduate Research Assistant, Data Science & AI

- Developed course content and hands-on labs for AI Tools and Apps for Business, teaching practical ML deployment, LLM workflows, Git, Docker, and responsible AI to MSBA students.
- Modernized 12+ modules in Predictive Analytics and Big Data, improving course materials and migrating content from R Markdown to Quarto.

University of New Hampshire, Durham, NH | BS in Information Systems & Business Analytics | Aug. 2021 – May 2025

NCAA Division I Cross Country & Track, Dean's List, America East Commissioner's Honor Roll

EXPERIENCE

Nassau Financial Group, Hartford, CT

June 2025 – August 2025

Data Engineer Intern (Solo Project Lead) | [View TPA Compliance Report](#)

- Built an R Shiny compliance analytics platform that prioritized review risk across 7 activity streams for 120K+ insurance policies, enabling faster issue triage for the compliance team.
- Designed a weighted compliance scoring framework across 8 insurance activity types and used 12-month trend analysis to reduce manual TPA review time by about 50%.

Sprague Operating Resources, Portsmouth, NH

June 2024 – August 2024

Business Intelligence & Data Intern

- Wrote SQL to extract and structure natural gas trading data, then modeled portfolio exposure metrics in Power BI to support the sales team.

SELECTED DATA SCIENCE PROJECTS

Airbnb: US Market Classifier & Paris Investment Study | [View Report](#)

March 2026 - May 2026

- Built a CatBoost classifier on 374K U.S. Airbnb listings with 45 features, 5-fold stratified cross-validation, and Optuna tuning across 40 trials; achieved 0.966 holdout AUC.
- Extended the analysis to a 93K-row Paris panel and found host operating scale remained a strong conversion signal after controlling for price features, with commercial hosts showing roughly 6x lower conversion efficiency than casual hosts.

Macroeconomic Anomaly Detection | [Live Demo](#)

January 2026 - March 2026

- Trained a PyTorch LSTM autoencoder on 7 FRED indicators using 12-month windows to detect multivariate macroeconomic anomalies through reconstruction error across major regime shifts.
- Used VECM cointegration analysis for feature design and calibrated thresholds on a pre-2007 baseline.

Loan Default Modeling and Drift Monitoring | [View Report](#)

December 2025

- Built a loan default prediction pipeline on 148K applications and achieved 0.92 ROC-AUC using a modeling-first workflow.
- Added PSI-based drift monitoring across 51 features to track post-deployment distribution shifts and support ongoing model risk review in a lending context.

Financial News Retrieval and Sentiment Analysis (RAG) | [Live Demo](#)

January 2026 - Present

- Built a retrieval and sentiment analysis system across 500+ financial articles to speed sector research workflows.
- A/B tested chunk size and retrieval window settings to optimize article context selection for analyst queries, then integrated the system into a live interface.

TECHNICAL SKILLS

Languages: Python (pandas, numpy, polars, PyTorch, scikit-learn), SQL, R (tidyverse, ggplot2, dplyr)

Data Science: classification, SHAP, feature engineering, model evaluation, cross-validation, time series analysis (VECM, ARIMA), causal inference, A/B testing, Bayesian optimization, drift monitoring

Applied ML & Tools: CatBoost, LightGBM, NLP, MLflow, Docker, Git/GitHub, Hugging Face, AWS, Quarto, Power BI